

EDC ASSIGNMENT 2

Question # 2

a) Using KVL at the base:

$$I_B = \frac{4.3V}{200k\Omega}$$

$$I_B = 21.5\mu A$$

For collector current

$$I_C = \frac{2V}{1k}$$

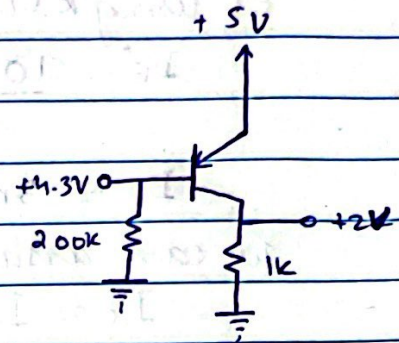
$$I_C = 2mA$$

Now finding current gain:

$$\beta = \frac{I_C}{I_B}$$

$$= \frac{2mA}{21.5\mu A}$$

$$\boxed{\beta = 93.02}$$



b) Using KVL at the base

$$I_B = \frac{4.3 - 2.3}{20k}$$

$$I_B = 100\mu A$$

same for collector current

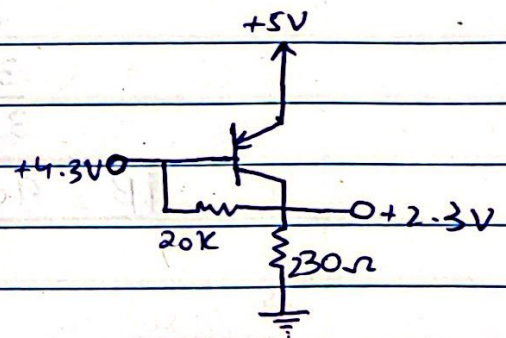
$$I_C = \frac{2.3}{0.230k}$$

$$I_C = 10mA$$

Now finding current gain.

$$\beta = \frac{I_C}{I_B} \Rightarrow \frac{10mA}{100\mu A}$$

$$\boxed{\beta = 100}$$



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c) Using KVL at the emitter

$$I_E = \frac{10 - 7}{1K}$$

$$I_E = 3mA$$

We can assume

$$I_C \approx I_E$$

$$I_C = 3mA$$

Now we can solve collector voltage

$$V_C = (1K)(3mA)$$

$$V_C = 3V$$

Now applying KVL at base

$$I_B = \frac{6.3 - 3}{100K}$$

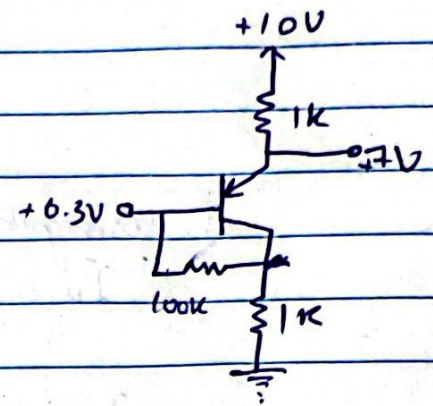
$$I_B = 33\mu A$$

Now finding current gain

$$\beta = \frac{I_C}{I_B}$$

$$= \frac{3mA}{33\mu A}$$

$$\beta = 90.91$$



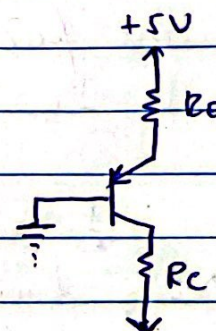
Question #8:

We have

$$I_E = 1mA, V_C = -1V$$

$$V_{EB1} = 0.64V, I_{E1} = 0.1mA$$

$$\beta = 100$$



solving for collector current

$$I_{C1} = \left(\frac{\beta}{\beta + 1} \right) I_{E1}$$

$$= \left(\frac{100}{100 + 1} \right) 0.1 \text{ mA}$$

$$I_{C1} = 0.099 \text{ mA}$$

Now solving for saturation current

$$I_{C1} = I_S \cdot e^{\frac{V_{BE1}}{V_T}}$$

$$I_S = \frac{0.099 \text{ mA}}{e^{\frac{0.64}{0.026}}}$$

$$I_S = 2.02 \text{ fA}$$

Finding collector current

$$I_C = \left(\frac{\beta}{\beta + 1} \right) I_E$$

$$= \left(\frac{100}{100 + 1} \right) 1 \text{ mA}$$

$$I_C = 0.99 \text{ mA}$$

Now finding V_{BE}

$$I_C = I_S \cdot e^{\frac{V_{BE}}{V_T}}$$

$$V_{BE} = 26 \text{ mV} \ln \left(\frac{0.99 \text{ mA}}{2.02 \text{ fA}} \right)$$

$$V_{BE} = 0.7 \text{ V}$$

For emitter voltage

$$V_E = V_B - V_{EB}$$

$$V_E = 0.7 \text{ V}$$

the formula for finding R_E is

$$R_E = \frac{5 \text{ V} - V_E}{I_E}$$

$$= 5V - 0.7V$$

$$1mA$$

$$R_E = 4.3K\Omega$$

And for R_C

$$R_C = V_C - (-5V)$$

$$I_C$$

$$= -1V - (-5V)$$

$$1mA$$

$$R_C = 4K\Omega$$

Question#1:

a) Calculating I_1

$$10000 = \frac{10.7 - 0.7}{I_1}$$

$$I_1$$

$$I_1 = \frac{10.7 - 0.7}{10000}$$

$$10000$$

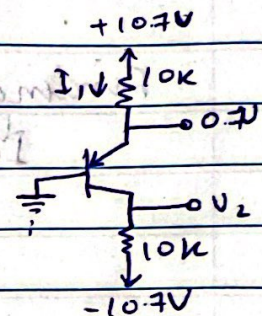
$$I_1 = 1mA$$

For V_2

$$10000 = \frac{V_2 - (-10.7)}{I_1}$$

$$V_2 = 10000 \times 1mA - 10.7$$

$$V_2 = -0.7V$$



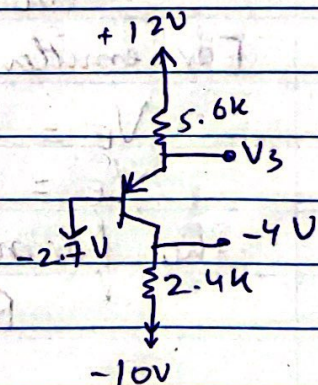
b) First finding I_C

$$2400 = \frac{-4 - (-10)}{I_C}$$

$$I_C$$

$$I_C = 2.5mA$$

Now V_3



$$5600 = \frac{12 - V_3}{2.5 \text{ m}}$$

$$V_3 = 12 - 5600 \times 2.5 \text{ m}$$

$$V_3 = -2 \text{ V}$$

c) we can calculate I_C

$$10 \text{ k} = \frac{0 - (-10)}{I_C}$$

$$I_C = \frac{10}{10 \text{ k}}$$

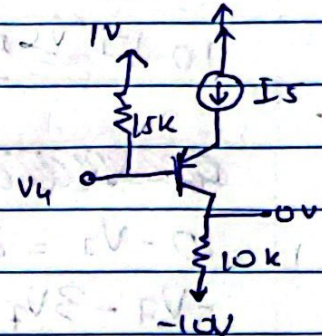
$$I_C = 1 \text{ mA}$$

$$I_S = I_C$$

$$I_S = 1 \text{ mA}$$

Also

$$V_4 = 1 \text{ V}$$



d) Firstly

$$I_C = I_E = I_B$$

So

$$I_C = \frac{10 - V_4}{15 \text{ k}} \rightarrow ①$$

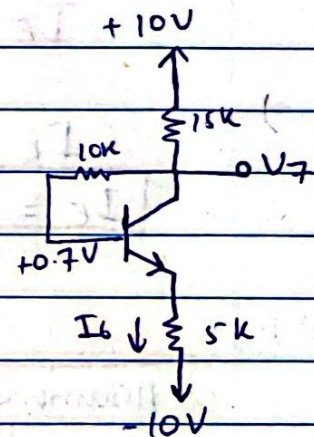
$$I_E = \frac{V_E - (-10)}{5 \text{ k}} \rightarrow ②$$

Also

$$V_{BE} = V_B - V_E$$

$$0.7 = V_7 - V_E$$

$$V_E = V_7 - 0.7$$



Now solving ① & ②

$$\frac{10 - V_7}{15K} = \frac{V_7 - 0.7 - (-10)}{5K}$$

$$\frac{10 - V_7}{15K} = \frac{9.3 + V_7}{5K}$$

$$10 - V_7 = \frac{9.3 + V_7}{5K} \times 15K$$

~~$$10 - V_7 = 27.9 + 3V_7$$~~

$$10 - V_7 = 27.9 + 3V_7$$

$$-V_7 - 3V_7 = 27.9 - 10$$

$$-4V_7 = 17.9$$

$$V_7 = \frac{17.9}{-4}$$

$$V_7 = -4.475$$

Now we can calculate I_E

$$I_E = \frac{10 - (-4.475)}{15K}$$

$$I_E = 0.965 \text{ mA}$$

$$I_E = I_C = 1.59 \text{ mA}$$

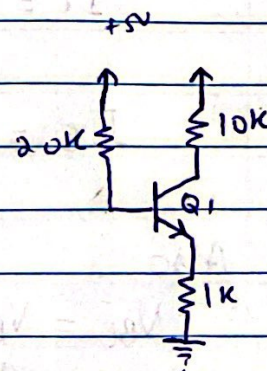
Question # 4:

a) Assuming

$$V_{CE} = 0.2 \text{ V} \text{ \& } V_{BE} = 0.5 \text{ V}$$

$$\beta_{\text{forced}} = \frac{I_C}{I_B}$$

Applying KVL from Base $\rightarrow$ emitter and from collector $\rightarrow$ emitter of Q_1 :



$$-5V + 20 \times 10^3 I_{B1} + V_{BE1} + I_{E1} (1 \times 10^3) = 0$$

$$-5V + 10 \times 10^3 I_{E1} + V_{BE1} + 1 \times 10^3 I_{E1} = 0$$

$$1 \times 10^3 (I_{E1}) = 5V - 10 \times 10^3 I_{B1} - V_{BE1}$$

$$-5V + 20 \times 10^3 I_{B1} + V_{BE1} + (5V - 10 \times 10^3 I_{C1} - V_{CE1}) = 0$$

$$20 \times 10^3 I_{B1} + 0.7V - 10 \times 10^3 I_{C1} - 0.2V = 0$$

$$20 \times 10^3 I_{B1} - 10 \times 10^3 I_{C1} = -0.5V$$

Expressing all currents in Q_1 in function of V_{C1} :

$$-5V + I_{B1} 20 \times 10^3 + V_{C1} - V_{BE1} = 0$$

$$I_{B1} = \frac{5V - V_{C1} - V_{BE1}}{20 \times 10^3}$$

$$= \frac{5V - V_{C1} - 0.5}{20 \times 10^3}$$

$$= \frac{4.5V - V_{C1}}{20 \times 10^3}$$

$$= \frac{4.5V - V_{C1}}{20 \times 10^3}$$

$$= \frac{4.5V - V_{C1}}{20 \times 10^3}$$

$$-V_{C1} + V_{CE1} + I_{E1} (10^3) = 0$$

$$I_{E1} = \frac{V_{C1} - V_{CE1}}{10^3}$$

$$= \frac{V_{C1} - 0.2V}{10^3}$$

$$= \frac{V_{C1} - 0.2V}{10^3}$$

$$= \frac{V_{C1} - 0.2V}{10^3}$$

$$-5V + V_{C1} + 10 \times 10^3 I_{C1} = 0$$

$$I_{C1} = \frac{5V - V_{C1}}{10 \times 10^3}$$

$$= \frac{5V - V_{C1}}{10 \times 10^3}$$

Applying KVL in Q_1 :

$$I_{B1} + I_{C1} + I_{E1} = 0$$

$$I_{B1} + I_{C1} = I_{E1}$$

$$\frac{4.5 - V_{C1}}{20} + \frac{5 - V_{C1}}{10} = \frac{V_{C1} - 0.2}{10}$$

$$4.5 - V_{C1} + 2(5 - V_{C1}) = 20(V_{C1} - 0.2)$$

$$4.5 - V_{C1} + 10 - 2V_{C1} = 20V_{C1} - 4$$

$$V_{C1} = 0.804V$$

verifying the assumption

$$\beta_{forced} < \beta_{min}$$

$$\beta_{forced} = \frac{I_{C1}}{I_{B1}}$$

$$= \frac{5V - V_{C1}}{10k}$$

$$\frac{4.5V - V_{C1}}{20k}$$

$$= 4.196 \times 10^{-4}$$

$$1.848 \times 10^{-4}$$

$$= 2.27$$

$$2.27 < 50 \quad (\text{Hence verified})$$

b) Expressing all currents in Q_2 in function of V_{C2} :

$$-5V + 1kI_{E2} + V_{CE2} + V_{C2} = 0$$

$$-5V + 1kI_{E2} + 0.2V + V_{C1} = 0$$

$$4.8V - V_{C1} = I_{E2}$$

$$1k$$

$$-(-5) + 10kI_{B2} + V_{BC} + V_{C2} = 0$$

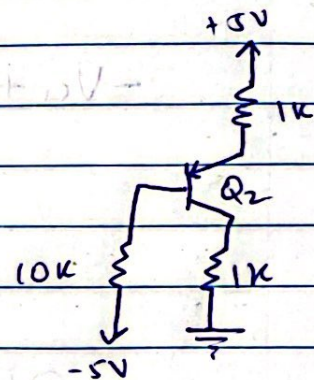
$$5 + 10kI_{B2} + 0.5 + V_{C2} = 0$$

$$\frac{-5.5 - V_{C2}}{10k} = I_{B2}$$

$$10k$$

$$-V_{C1} + I_{C2}(1k) = 0$$

$$\frac{V_{C1}}{10^3} = I_{C2}$$



Applying KCL in Q_2

$$I_{B2} + I_{E2} = I_{C2}$$

$$-\frac{(5.5 + V_{C2})}{10k} + \frac{4.8 - V_{C2}}{1k} = \frac{V_{C2}}{51k}$$

$$-(5.5 + V_{C2}) + 10(4.8 - V_{C2}) = 10V_{C2}$$

$$-5.5 - V_{C2} + 48 - 10V_{C2} - 10V_{C2} = 0$$

$$\frac{42.5}{21} = V_{C2}$$

$$V_{C2} = 2.02V$$

Verifying the assumption

$$\beta_{\text{forced}} = \frac{I_{C2}}{I_{B2}}$$

$$= \frac{V_{C2}/1k}{(5.5 + V_{C2})/10k}$$

$$= \frac{2.02}{5.5 + 2.02/0.752}$$

$$\beta_{\text{forced}} = 2.686$$

$$\beta_{\text{forced}} < \beta_{\text{min}}$$

$$2.686 < 50$$

(Hence proved)

Question # 6:

$$\beta = 50$$

$$V_E = 5V$$

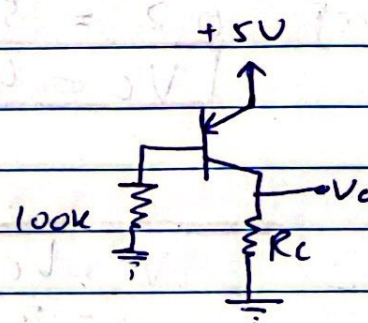
calculating V_B

$$V_{EB} = 0.7$$

$$V_E - V_B = 0.7$$

$$5 - V_B = 0.7$$

$$V_B = 4.3V$$



finding I_B

$$I_B = \frac{V_B}{R_B} \Rightarrow \frac{4.3}{100k}$$

$$I_B = 4.3 \times 10^{-5} A$$

Finding I_C

$$I_C = \beta \cdot I_B \Rightarrow 50 \times 4.3 \times 10^{-5}$$

$$I_C = 2.15 mA$$

Finding R_C

$$R_C = \frac{V_C}{I_C} \Rightarrow \frac{3}{2.15m}$$

$$R_C = 1395 \Omega$$

$$\beta = 100$$

I_C becomes

$$I_C = \beta \cdot I_B \Rightarrow 100 \times 4.3 \times 10^{-5}$$

$$I_C = 4.3 mA$$

$$V_C = R_C I_C$$

$$= (1395)(4.3m)$$

$$V_C = 6V$$

$$V_{EC} = V_E - V_C$$

$$0.2 = 5 - V_C$$

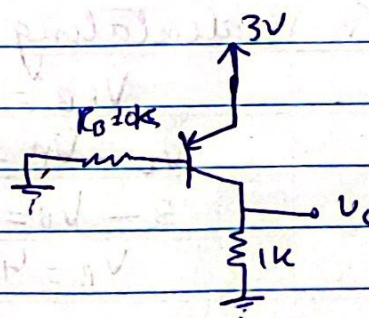
$$V_C = 4.8V$$

Question # 10:

$$V_C = V_E - V_{EC}$$

$$= 3 - 0.2$$

$$V_C = 2.8V$$



$$I_C = \frac{V_C}{R_C} \Rightarrow \frac{2.8}{1000} \Rightarrow 2.8 \times 10^{-3} \text{ A}$$

Assuming $V_{EB} = 0.7$

$$V_B = V_E - V_{EB} = 3 - 0.7 = 2.3 \text{ V}$$

$$I_B = \frac{V_B}{R_B} = \frac{2.3}{10\text{K}} \Rightarrow 2.3 \times 10^{-4} \text{ A}$$

$$\beta_{\text{forced}} = \frac{I_C}{I_B} \Rightarrow \frac{2.8 \times 10^{-3}}{2.3 \times 10^{-4}}$$

$$\beta_{\text{forced}} = 12.1$$

Assuming $V_{EC} = 0.3 \text{ V}$

$$V_C = V_E - V_{EC} \Rightarrow 3 - 0.3$$

$$V_C = 2.7 \text{ V}$$

$$I_C = \frac{V_C}{R_C} \Rightarrow \frac{2.7}{1000}$$

$$I_C = 2.7 \text{ mA}$$

As β is 50

$$I_B = \frac{I_C}{\beta} \Rightarrow \frac{2.7 \text{ mA}}{50} \Rightarrow 5.4 \times 10^{-5}$$

Now R_B is

$$R_B = \frac{V_B}{I_B} = \frac{2.3}{5.4 \times 10^{-5}}$$

$$R_B = 42593 \Omega$$

Question # 7:

$$V_{CE} = V_{CC} - I_C R_C$$

$$V_{CE} = 10 - 1 \text{ mA} \times 1\text{K} \Rightarrow 9 \text{ V} \quad (I_C = 1 \text{ mA})$$

$$V_{CE} = 10 - 5\text{mA} \times 1\text{k}\Omega = 5\text{V} \quad (I_C = 5\text{mA})$$

$$V_{CE} = 10 - 8\text{mA} \times 1\text{k}\Omega = 2\text{V} \quad (I_C = 8\text{mA})$$

$$V_{CE} = 10 - 9\text{mA} \times 1\text{k}\Omega = 1\text{V} \quad (I_C = 9\text{mA})$$

Voltage gain is

$$\frac{V_o}{V_i} = \beta \frac{R_C}{R_B} \Rightarrow \frac{I_C R_C}{I_B R_B} = \frac{V_o}{V_i}$$

$$I_B R_B \approx V_{BB} \text{ as}$$

$V_{BE} \approx 0.7$ which is small as compared to V_{BB}

$$\text{For } I_C = 1\text{mA} \Rightarrow A_V = \frac{1\text{mA} \times 1\text{k}\Omega}{V_{BB}} = \frac{1}{V_{BB}}$$

$$\text{For } I_C = 5\text{mA} \Rightarrow A_V = \frac{5\text{mA} \times 1\text{k}\Omega}{V_{BB}} = \frac{5}{V_{BB}}$$

$$\text{For } I_C = 8\text{mA} \Rightarrow A_V = \frac{8\text{mA} \times 1\text{k}\Omega}{V_{BB}} = \frac{8}{V_{BB}}$$

$$\text{For } I_C = 9\text{mA} \Rightarrow A_V = \frac{9\text{mA} \times 1\text{k}\Omega}{V_{BB}} = \frac{9}{V_{BB}}$$

Sr #	I_C	V_{CE}	A_V	+ve signal swing	-ve signal swing
1	1	9	$1/V_{BB}$	$V_i \times 1/V_{BB}$	$-V_i \times 1/V_{BB}$
2	5	5	$5/V_{BB}$	$V_i \times 5/V_{BB}$	$-V_i \times 5/V_{BB}$
3	8	2	$8/V_{BB}$	$V_i \times 8/V_{BB}$	$-V_i \times 8/V_{BB}$
4	9	1	$9/V_{BB}$	$V_i \times 9/V_{BB}$	$-V_i \times 9/V_{BB}$

Question #5

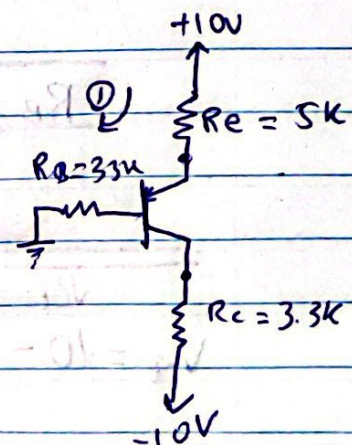
$$V_{BE} = 0.7\text{V}$$

$$\beta = 50$$

KVL at loop ①

$$10 = 5 I_E + V_{BE} + 33 I_B$$

$$10 - V_{BE} = (5(\beta + 1) I_B + 33 I_B)$$



$$255 I_B + 33 I_B = 10 - 0.7$$

$$I_B = 0.0322 \text{ mA}$$

$$I_C = \beta I_B$$

$$= (50)(0.0322)$$

$$I_C = 1.6145 \text{ mA}$$

$$I_E = (1 + \beta) I_B =$$

$$I_E = 1.6468 \text{ mA}$$

$$a) V_B = I_B R_B = 0.0322 \times 33 \Rightarrow 1.0626 \text{ V}$$

$$V_C = I_C R_C = 1.6145 \times 3.3 \Rightarrow 5.3278 \text{ V}$$

$$V_E = I_E R_E = 1.6468 \times 5 \Rightarrow 8.234 \text{ V}$$

$$V_B = 1.0626 \text{ V}$$

$$V_C = 5.3278 \text{ V}$$

$$V_E = 8.234 \text{ V}$$

$$b) V_{EC} = 10 - V_E - V_C$$

$$= 10 - 8.234 - 5.327$$

$$V_{EC} = -3.561 \text{ V}$$

$$(V_{EB} = 0.7 \text{ V})$$

$$V_{EC} < V_E \Rightarrow \text{BJT operates in saturation region}$$

$$c) \text{ Change } R_C \text{ from } 3.3 \text{ k}\Omega \text{ to } 33 \text{ k}\Omega$$

$$V_C = R_C I_C$$

$$= 33 \times 1.6145$$

$$V_C = 53.27$$

$$V_{EC} = 10 - 53.27 - 8.234$$

$$V_{EC} = -51.512 \text{ V}$$

$$V_{EB} = 0.7 \text{ V}$$

$$V_{EC} < V_E \Rightarrow \text{BJT operates in saturation region}$$

d) When BJT is in active region $V_{EC} = 0.2V$

$$V_{EC} = 10 - R_E I_E - R_C I_C$$

$$0.2 = 10 - 8.234 - R_C \times 1.614$$

$$R_C = \frac{1.766 - 0.2}{1.614}$$

$$R_C = 970 \Omega$$

Question #9

At Q_1

$$R = \frac{5 - 0}{1m} \Rightarrow 5K \Omega$$

$$V_{BC} = V_B - V_E = 0.2$$

$$= -0 - V_E = 0.2$$

$$R_P = \frac{V_E}{I_E} = \frac{0.2}{1} \Rightarrow 0.2K \Omega$$

At Q_2

$$V_S = -2V$$

$$\frac{-2 + 5}{1m} \Rightarrow R_4 = 3K \Omega$$

$$V_{BE} = -0.2, V_C = 0.2$$

$$R_3 = \frac{5 - 0.2}{1m} = 4.3K \Omega$$

At Q_3

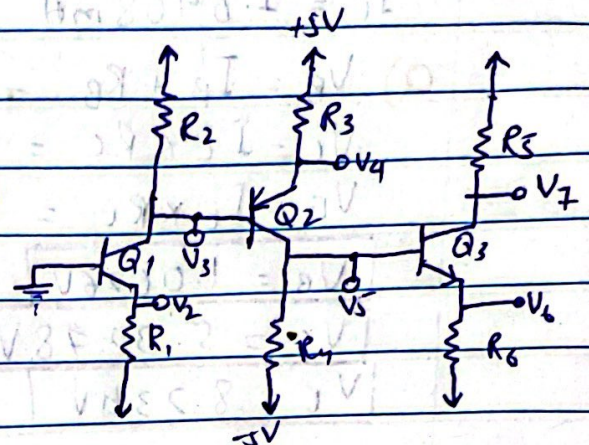
$$V_{BE} = 0.2$$

$$V_B - V_E = 0.2 \Rightarrow -2 - V_E = 0.2$$

$$V_E = -1.3V$$

$$R_6 = \frac{V_E}{I_E} \Rightarrow \frac{-1.3}{2} \Rightarrow 0.65K \Omega$$

$$R_5 = \frac{5 - V_2}{2m} \Rightarrow \frac{5 - 1}{2m} \Rightarrow \frac{4}{2m}$$



$$R_5 = 2k$$

$$\text{Assume } \beta = 100$$

$$V_3 = 5 - 1(5.1) \Rightarrow -0.1V$$

$$V_4 = 5 - (1.001)(4.31) \Rightarrow 0.69V$$

$$V_5 = -5 + (1)(3) \Rightarrow -2V$$

$$V_5 - V_6 \Rightarrow 0.2 \quad (V_6 = 2.2)$$

$$V_2 = 5 - 1(2) \Rightarrow 3$$

$$I_c = I_E$$

$$(1 + 1/\beta)$$

$$= 1 \left(1 + \frac{1}{10} \right)$$

$$= 1(1 + 0.001)$$

$$I_{c1} = 1.001$$

$$I_{c2} = 1.001$$

$$I_{c3} = 2.002$$